

P2559R1

Plan for Concurrency Technical Specification Version 2

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Source:

[GitHub](#)

Issue Tracking:

[GitHub](#)

Project:

ISO/IEC JTC1/SC22/WG21 14882: Programming Language — C++

Audience:

WG21

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§ 1. Introduction

We've been slowly filling the Concurrency Technical Specification (TS) version 2 for approximately six years. It's time to move this work to publication so we can get field experience with the contents of the TS.

§ 2. Scope

The following content is already in the Concurrency TS v2 Working Draft [\[N4895\]](#):

- [\[P1121R3\]](#): Hazard Pointers
- [\[P1122R4\]](#): Read-Copy-Update (RCU) Data Structures

We should apply the following papers to the Concurrency TS v2 Working Draft [\[N4895\]](#) and then publish it:

- [\[P0290R2\]](#): `synchronized_value`
- [\[P1478R7\]](#): Byte-wise Atomic `memcpy`
- [\[P2396R0\]](#): Concurrency TS 2 Fixes
- [\[P1202R4\]](#): Asymmetric Fences

§ 3. Timeline

- 2022-2H: Complete Concurrency and Library Evolution review of all Concurrency TS v2 content.
- 2023-1H: Complete Library review of all Concurrency TS v2 content.
- 2023-2H: Publish the Concurrency TS v2.

§ 4. Changelog

- R0: Initial proposal.
- R1: Updated to the scope approved by the Concurrency and Parallelism Study Group and Library Evolution.

§ References

§ Informative References

[N4895]

Michael Wong. *Working Draft, Extensions to C++ for Concurrency Version 2*. 15 August 2021. URL: <https://wg21.link/n4895>

[P0290R2]

Anthony Williams. *apply() for synchronized_value<T>*. 2 March 2017. URL: <https://wg21.link/p0290r2>

[P1121R3]

Maged M. Michael, Michael Wong, Paul McKenney, Geoffrey Romer, Andrew Hunter, Arthur O'Dwyer, D. S. Hollman, JF Bastien, Hans Boehm, David Goldblatt, Frank Birbacher, Mathias Stearn, Jens Maurer. *Hazard Pointers: Proposed Interface and Wording for Concurrency TS 2*. 9 April 2021. URL: <https://wg21.link/p1121r3>

[P1122R4]

Paul E. McKenney, Paul McKenney, Michael Wong, Maged M. Michael, Geoffrey Romer, Andrew Hunter, Arthur O'Dwyer, Daisy Hollman, JF Bastien, Hans Boehm, David Goldblatt, Frank Birbacher, Erik Rigtorp, Tomasz Kamiński, and Jens Maurer. *Proposed Wording for Concurrent Data Structures: Read-Copy-Update (RCU)*. 14 May 2021. URL: <https://wg21.link/p1122r4>

[P1202R4]

David Goldblatt. *Asymmetric Fences*. 15 February 2022. URL: <https://wg21.link/p1202r4>

[P1478R7]

Hans Boehm. *Byte-wise atomic memcpy*. 5 February 2022. URL: <https://wg21.link/p1478r7>

[P2396R0]

David Goldblatt. *Concurrency TS 2 fixes*. 14 June 2021. URL: <https://wg21.link/p2396r0>